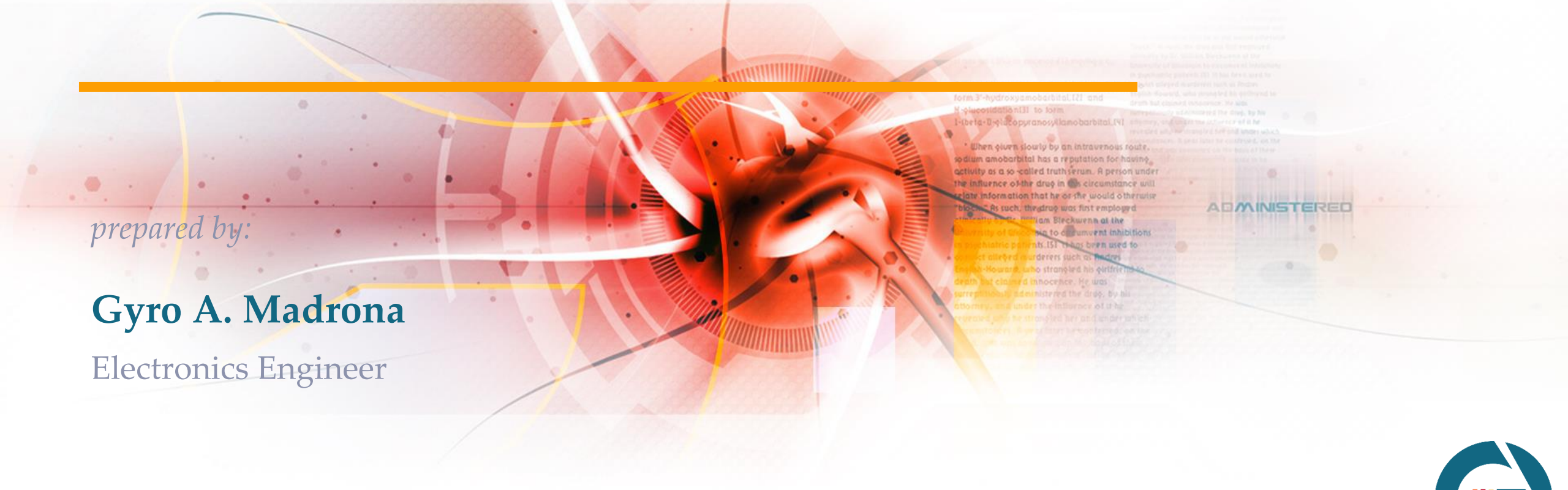
THE SPHERE

SOLID MENSURATION



prepared by:

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GRADING SYSTEM

FINAL GRADE

30%

PRELIM

30%

MIDTERM

40%

ENDTERM

TERM GRADE

70%

WEEKLY

30%

EXAM

70%

LONG QUIZ

30%

SEATWORK

PENALTY

0

ABSENT

3

45-min LATE

7 max

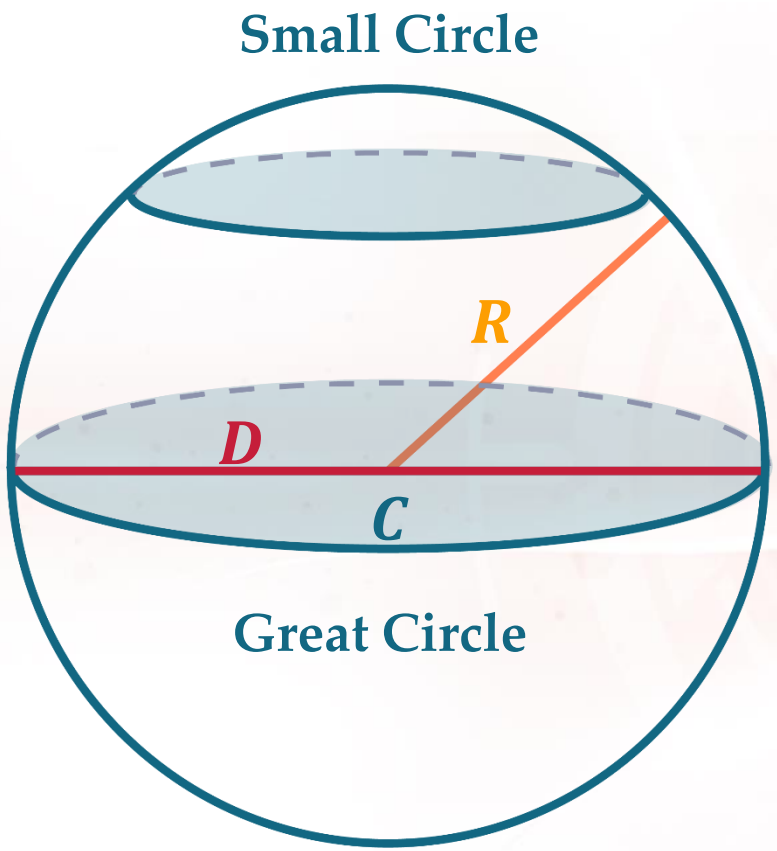
15-min LATE

TOPIC OUTLINE

Properties of a Sphere

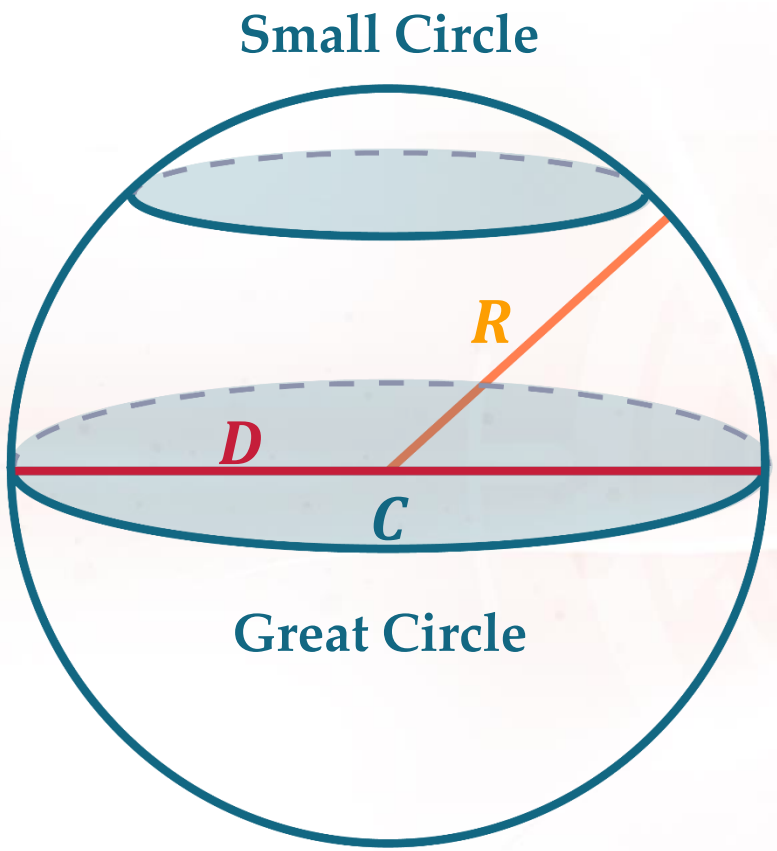
- Surface Area
- Volume

PROPERTIES OF A SPHERE



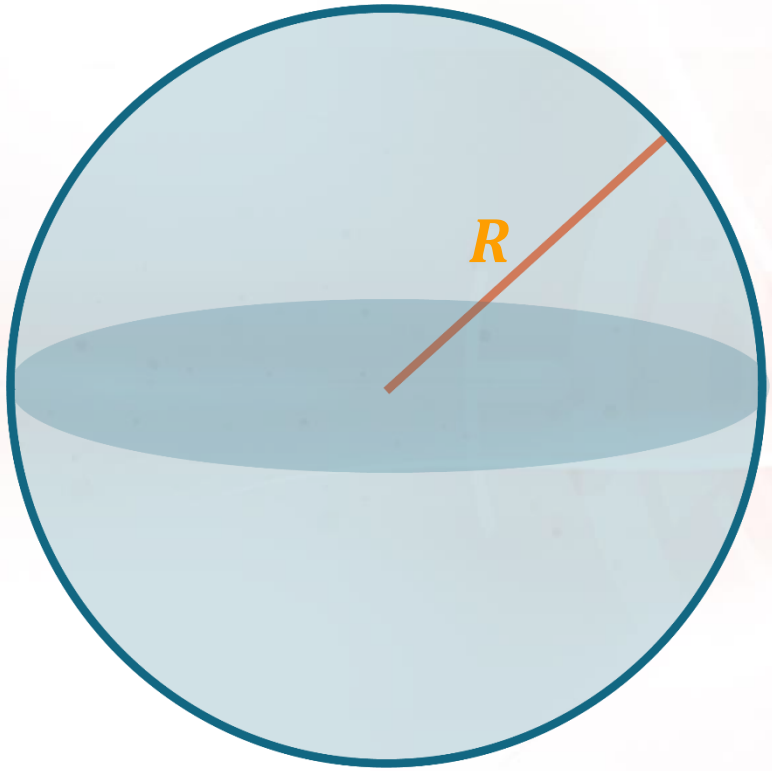
A **sphere** is a solid bounded by a closed surface every point of which is **equidistant,** ***R*** from a fixed point called the **center, *C***.

PROPERTIES



1. Every plane section of a sphere is a circle. If the plane contains a diameter of the sphere, the section is a **great circle**; otherwise, the section is a **small circle**.
2. The **axis** of a sphere is the **diameter, *D*** of the sphere perpendicular to the plane of the circle.
3. The **radius, *R*** of a great circle is equal to the radius of the sphere.

AREA OF A SPHERE



The Surface Area, **S**

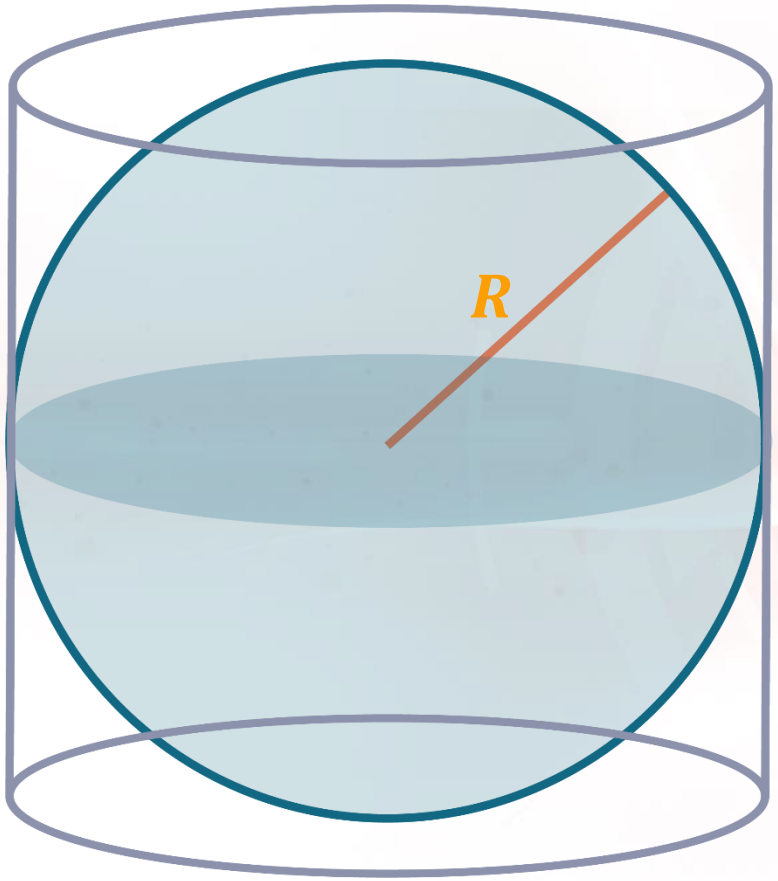
$$\boldsymbol{S} = area\ of\ 4\ great\ circles$$

$$\boldsymbol{S = 4\pi r^2}$$

where,

r = radius of the sphere

Consider a **solid sphere** with a **radius** of **5** cm. Calculate the **cost** of painting it, assuming that the cost of paint is **$0.08 per square centimeter**.



The Volume, **V**

$$V = \frac{4}{3}Bh$$

$$V = \frac{4}{3}\pi r^3$$

where,

r = radius of the sphere

Find the **weight** of a **snowball 4** ft. in **diameter** if the wet compact snow of which this ball is made weighs **30 lb. per cu. ft**.

An **iron ball 4** in. in **diameter** weighs **9 lb**.

Find the **weight** of an **iron shell 2 in. thick**

whose **external** diameter is **20 in.**

SEATWORK

1. The surface of a **hemispherical** dome whose **diameter** is **36** ft. is to be covered with gold leaf which costs **15 cents per square inch**. What must be **paid** for the gold leaf?

2. A **spherical** wooden ball **1** ft. in **diameter** is used as a float for a boat. If the ball sinks to a **depth** of **7 in**. , find the **area** of the **wetted** surface. *Hint: Zone of a Sphere*

3. A wooden **sphere** of **radius 15** in. rests in a circular hole in a board. The radius of the **hole** is **5** in. How **far** below the upper surface of the board does the sphere extend?